

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record its brain responses with our lab's probes. We are planning the sound setup. SONIC tested awake sheep; our mouse state and tone settings remain open. (Methods 6.6, p.13)

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- **SONIC:** initially explored 1.4–32 kHz; benchmark tones were 10 ms sine waves with 2 ms fades at each end. (Methods 6.1–6.2, p.11)
- **SONIC:** random frequencies, balanced within each block, with no silence between tones. (Methods 6.1, p.11)
- Save the sequence before playback, as SONIC did. (Methods 6.2, p.11)

2. Set up the speaker

- Plan as though we need to assemble the whole sound setup.
- **SONIC:** NI USB-6361 output at 192 kHz, TDT SA1 amplifier, and MF1 free-field speaker; about 80 dB SPL at sheep-ear distance. (Methods 6.2, p.11)
- **Our plan:** use the proposed shopping list on the next page, keep speaker placement fixed, and measure each tone with a calibrated microphone at the mouse-ear position.

3. Keep everything on one timeline

- Use our lab's probes. Confirm the recording system's timing input so it can record the tone markers alongside the neural signals.
- **SONIC:** recorded electrical sound output and trial-start markers; shared clock references aligned these with Connexus brain recordings. (Methods 6.2, p.11)
- **Our additions:** synchronize a microphone recording to check actual sound arrival. Align video frame times too if we add behavior video.

Reference: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1. Page numbers above refer to the printed manuscript. Mouse choices and additions are proposals.

Sound setup: shopping list

Use our lab's probes

USD

Budget for one of each item or group below. **Est.** means our rough allowance, not a confirmed supplier price.

What we get	What it does	Budget
1 TDT MF2-M speaker kit	Plays the tones.	\$1,500 est.
1 TDT SA1 amplifier	Drives the speaker.	\$2,000 est.
1 TDT ZB1PS power unit	Powers the amplifier.	\$750 est.
1 NI USB-6361	Controls tones and timing.	\$3,030
1 complete Avisoft 116Hm kit	Records sound at the ear position.	\$8,160
Calibration support	Checks the sound levels.	\$1,500 est.
1 computer, monitor, and storage set	Runs playback and saves data.	\$2,000 est.
Cables, timing connections, and mounts	Connects equipment and fixes its position.	\$1,000 est.
Sound setup subtotal	Rounded budget	\$20,000

What is already included?

- The Avisoft kit includes the microphone, preamp, sound recorder, cable, stand, and software. Buy **one kit**.
- The MF2-M kit includes a stand and connection accessories. No extra speaker or microphone is needed for this budget.

What do we already have?

- Our lab provides the probes. Neural hardware and animal preparation are outside this sound budget.
- Reuse a suitable lab computer and storage if available; then remove the \$2,000 allowance above.

Allow for extras

- Keep **\$2,500 reserve** for price changes, shipping, and tax: sound budget **about \$22,500**.
- Optional behavior camera with accessories: **about \$1,100 extra**.

Before buying: Confirm the lab recorder's timing input. The microphone's stated band starts at 2 kHz; lower tones need a different choice. Calibration and timing need checking. A soundproof box is not required.

Sources, October 8, 2026: TDT kit, amplifier, power, NI, Avisoft kit, prices. SONIC used MF1; MF2 is our current candidate. Estimates/alternatives are detailed in the repository. This mouse draft is separate from Rice's sedated-sheep plan.